

Prem Raichura

Full stack developer | AI/ML Researcher | DevOps & MLOps
| 1 x Conference Publication

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SUMMARY

A Full-Stack & AI Developer with a strong focus on building intelligent, scalable, and user-centric solutions. I enjoy transforming complex ideas into clean, functional systems — from robust backend architectures to intuitive frontend interfaces. My experience spans across AI/ML-driven applications, deep learning research, and web development using technologies such as FastAPI, React, PHP, Python, and MySQL.

EXPERIENCE

12/2025 – 04/2026

DevOps & Backend Developer Intern

Institute for Plasma Research (IPR), Gandhinagar | Certificate

- Conducted a comparative analysis of ERPNext, Odoo, and Dolibarr based on architecture, scalability, module availability, hardware requirements, and integration capabilities to identify the optimal ERP solution.
- Deployed and containerized ERPNext on Ubuntu using Docker and MariaDB, ensuring portability, scalability, and streamlined system configuration.
- Implemented HRMS, attendance tracking, shift management, automated email notifications, and role-based access control (RBAC) modules tailored to organizational requirements.
- Developed real-time attendance dashboards with employee check-in/check-out tracking, shift assignment workflows, and monitoring analytics.
- Resolved deployment and database configuration challenges involving Docker networking and MariaDB permissions, improving system reliability and maintainability.

RESEARCH & PUBLICATIONS

IEEE Xplore

TrafficEye: Intelligent Traffic Optimization Using Deep Learning Approach

Published | Doi Link

Built and evaluated an AI-driven traffic optimization system using YOLOv5s on the IITM-HeTra dataset for real-time vehicle detection and analysis. Contributed to model development, experimentation, and results validation, leading to successful publication in IEEE AIMV 2025.

Accepted

SecureSamvaad : A Multimodal Fusion Approach for Real-Time Telecom Fraud Detection

International Conference on Converging Intelligence - 2026.

Proposed a multimodal AI framework integrating diverse data sources for real-time telecom fraud detection. Developed machine learning models and fusion strategies to improve fraud identification accuracy and reduce false positives.

Accepted

SecurePark : Real-Time Edge AI and IoT Based Vehicle Parking Ecosystem

International Conference on Converging Intelligence - 2026.

Designed a smart parking ecosystem leveraging Edge AI and IoT technologies for real-time vehicle detection, parking space monitoring, and intelligent parking management. Contributed to system architecture, model development, and deployment strategy.

Under Review

PharmaADR-AI: A Pharmacovigilance-Driven Explainable Ensemble Machine Learning Framework for Opioid Adverse Drug Reaction Prediction

Elsevier: Computational Toxicology Journal.

Developed an explainable ensemble machine learning framework for pharmacovigilance-driven prediction of opioid adverse drug reactions (ADRs). Focused on enhancing drug safety monitoring through interpretable AI models and comprehensive adverse event analysis.

Under Review

From Theory to Practice: A Survey and Case-Based Analysis of Face-Swapping Deepfake Detection Models

MDPI: MultiMedia Tools and Applications Journal.

Conducted a comprehensive survey and case-based analysis of state-of-the-art face-swapping deepfake detection techniques. Evaluated existing approaches, datasets, challenges, and future research directions in deepfake detection.

EDUCATION

2024 – 2027

LDRP Institute of Technology and Research, Gandhinagar

B.E. Computer Engineering – CPI: 7.22/10 (Till 6-sem)
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Artificial Intelligence, Machine Learning, Deep Learning, Software Engineering.

2020 – 2023

Government Polytechnic, Porbandar

Diploma in Computer Engineering – CGPA: 7.09/10

PROJECTS

June 2026

ParkVue – AI-Driven Parking Hotspot Prediction and Traffic Congestion Mitigation System

GitHub | Live Demo

- Developed an end-to-end AI-powered platform to detect parking violation hotspots and reduce traffic congestion using real-time violation data and Uber H3 hexagonal spatial indexing
- Built a supervised machine learning pipeline to forecast parking hotspot severity of 24 hours and 48 hours with the achieved precision of 71%, enabling proactive traffic management and enforcement planning.
- Designed an officer-facing mobile application with optimized patrol task allocation and integrated Explainable AI (XAI) for transparent and interpretable decision-making.
- Tech Stack: React, TypeScript, Vite, React-Native, Tailwind CSS, AI APIs.

June 2026	VoiceScript Studio – AI-Powered Voice-to-Content Platform GitHub Live Demo
	- Developed an AI-powered speech processing platform that converts audio recordings into structured text, enabling efficient voice-to-content generation workflows. - Implemented support for direct Google Drive audio links and file uploads, automating audio retrieval, transcription, and content extraction through intelligent processing pipelines. - Designed a responsive and intuitive user interface with real-time processing feedback, enhancing accessibility and productivity for content creators and professionals. - Tech Stack: React, TypeScript, Vite, Tailwind CSS, AI APIs, Speech-to-Text Processing.
June 2026	PortOS – Personal Portfolio Operating System GitHub Live Demo
	- Engineered a scalable Portfolio Operating System with a centralized admin dashboard for managing projects, work experience, skills, certifications, and academic achievements. - Designed and integrated secure REST APIs, authentication mechanisms, and database operations to support seamless content management and automated portfolio updates. - Tech Stack: React, TypeScript, Vite, Tailwind CSS, Node.js, Express.js, MongoDB.
November 2025	Traffic Density & Detection Web App GitHub Live Demo
	- Developed a real-time traffic monitoring web application using React, Vite, FastAPI, and YOLOv5s trained on the IITM-HeTra dataset for vehicle detection and traffic density estimation. - Achieved 71% detection accuracy with 1.4 ms inference time through an optimized AI/ML pipeline, enabling efficient real-time traffic analysis and visualization. - Tech Stack: React + Vite, FastAPI, YOLOv5s, AI/ML.
October 2025	WealthEquity Live Website
	- Developed the official WealthEquity website using React, Vite, and Tailwind CSS with a fully responsive and animated user interface. - Integrated Email.js for automated inquiry handling, improving lead generation and contributing to an increase of 5+ clients through enhanced digital engagement. - Tech Stack: React + Vite, Tailwind CSS, Email.js.
September 2025	Prompt Injection Prevention Github Link
	- Developed a hybrid AI safety framework combining rule-based filtering and a Vision Transformer (ViT) classifier to detect malicious and unsafe prompts. - Achieved 98.32% classification accuracy by training and optimizing the model on labeled prompt datasets, enhancing LLM security and reliability. - Tech Stack: Python, PyTorch, Vision Transformer (ViT), AI/ML.

Achievements

2026	Finalist – InnovAltion Hackathon 2026 GitHub Link LinkedIn Post
	Built EDITH, an AI-powered medical imaging system for early disease detection, and advanced to the final round of the national-level hackathon.
2026	Finalist – ANRF AISEHack 2026 LinkedIn Post
	Fine-tuned IBM's Prithvi-100M geospatial foundation model for flood detection from satellite imagery, achieving a benchmark score of 0.703.
2025	Research Project Grant Recipient – KSV LinkedIn Post
	Secured ₹69,017 in competitive research funding for SecurePark, an Edge AI, Computer Vision, and IoT-based smart campus security and parking solution.
2025	Research Publication Grant Recipient – KSV LinkedIn Post
	Awarded a publication grant for presenting "TrafficEye: Intelligent Traffic Optimization Using Deep Learning Approach" at an international conference.
2024	Finalist – DAU Tic Tae Toe Hackathon GitHub Link
	Developed EDUMATE, a digital learning platform, and advanced to the finalist stage among participating teams.

Leadership & Volunteer Experience

May 2026 - Present	Reasearch Outreach Commitee Member IEEE KSV Student Branch
	Selected as a member of the Research Outreach Committee under the IEEE Student Branch. Engaging with research-oriented initiatives, technical activities, and academic outreach programs within the IEEE community.

Skills

- AI / ML & Research:** Deep Learning, Machine Learning, Computer Vision, Vision Transformers (ViT), Object Detection, Image Classification, Model Fine-Tuning & Model Evaluation.
- AI Frameworks & Libraries:** TensorFlow, PyTorch, Pandas, NumPy, OpenCV, Keras.
- Backend Development:** NodeJS, ExpressJS, Python, FastAPI, Django, Flask, PHP, Laravel, RESTful APIs, Authentication & Authorization, API Design, Database Integration.
- Frontend & Mobile Development:** React.js, Vite, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, React Native, Android Development.
- Programming Languages:** Python, PHP, JavaScript, TypeScript, Java, SQL, C, C++.

- **Databases & Cloud Services:** MySQL, Firebase, Supabase, PostgreSQL, Database Design, Query Optimization.
- **DevOps & Tools:** Git, GitHub, Docker, Postman, Linux, CI/CD Fundamentals, Deployment on Vercel, Render & Cloud Platforms.
- **Soft Skills:** Research Writing, Technical Documentation, Problem Solving, Team Collaboration.

CERTIFICATIONS

NPTEL	Python for Data Science (Elite)	Certificate Link
Certification	Basic & Advanced Python	Certificate Link